

SYSTEMS DESIGN

Systems Analysis & Design

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Outline

1 Requirements Engineering

2 Design & Process



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2 Design & Process



Stakeholders Vs. Shareholders

- **Stakeholders** are individuals or groups who have an interest in the success of a project.
- **Stakeholders** can be internal or external to a company. For example, customers, employees, suppliers, and regulators are external stakeholders.
- **Shareholders** are individuals or groups who have an ownership interest in a company.
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Requirements

big functionality

- **Requirements** are statements that describe the features, functions, and constraints of a system.
- **Requirements** are used to communicate the needs of stakeholders to developers.

- Requirements are used to guide the design, development, and

Feature: enrollments, bills & taxes

function: .doc 2 .pdf / apply filter

constraint: roles-permissions / time
/ cost

needs



Requirements

- **Requirements** are statements that describe the features, functions, and constraints of a system.
- **Requirements** are used to communicate the needs of stakeholders to developers.
- **Requirements** are used to guide the design, development, and testing of a system.

acceptance

contract

Final user



User Stories → non-tech

- **User stories** are **short, simple** descriptions of a **feature** or **function** of a **system**.
- They are **written** from the **perspective** of the **user** and **describe** what the **user** wants to **achieve**.
- They are **used** to **capture** the **requirements** of a **system** in a **simple** and **understandable** way.

client tech team



User Stories: Format Example

User Story

Title: Word to PDF	Priority: High Medium Low	Estimate: ~1h ~2h
User Story: As a [description of user], I want [functionality] so that [benefit].		
Acceptance Criteria: Given [how things begin] When [action taken] Then [outcome of taking action]		

As a professor
I want to be able
so that we can have online c...

① Given guests delivery
when I choose send and.
then are done

all acceptance criteria




What is Requirements Engineering?

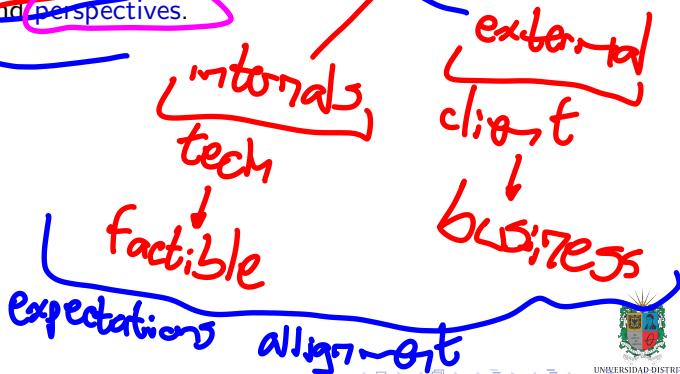
- **Requirements engineering** is the process of eliciting, analyzing, specifying, validating, and managing the requirements of a system.
- It is a critical activity in the systems development lifecycle that ensures that the system meets the needs of its users.
- It is a collaborative process that involves stakeholders from different backgrounds and perspectives.

كله لوج
لو



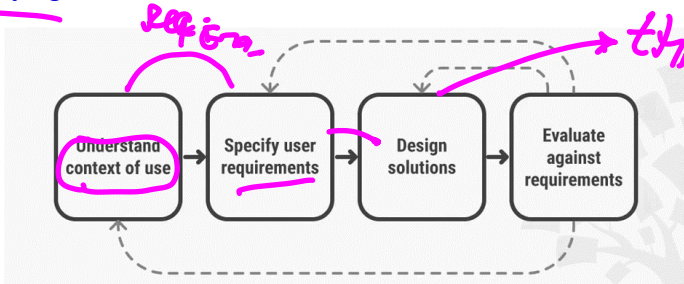
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User-Centered Design (UCD)

- **User-centered design** (UCD) is an iterative design process that focuses on understanding the needs, preferences, and behaviors of users. *Final Users*
- UCD is a collaborative process that involves users in the design and development of a system.
- UCD is used to create systems that are usable, efficient, and satisfying to users.



Requirements Engineering Process

The **requirements engineering** process consists of the following activities:

- Gathering requirements.
- Analyzing requirements.
- Validating requirements.
- Verifying requirements.
- Documenting requirements.
- Managing requirements.
- Communicating requirements.



Gathering Requirements

- **Gathering** requirements is the process of collecting and documenting the needs of stakeholders.
- It involves interviewing stakeholders, conducting surveys, and observing users to understand their requirements.
- It is a collaborative process that involves stakeholders from different backgrounds and perspectives.

Handwritten notes in pink:
- A circle around "needs" with an arrow pointing to "externals".
- The word "externals" written above the second bullet point.
- The phrase "Business Understanding" written diagonally across the third bullet point.



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Clients are not always right

Dear Santa,
How are you? I'm good.
Here is what I want for
Christmas.

✗

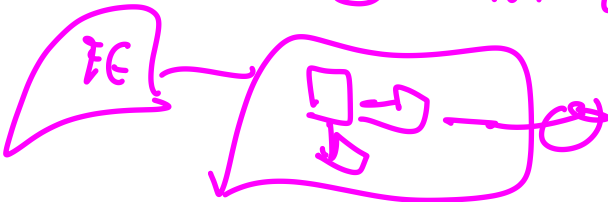
http://www.amazon.com/gp/product/B0032HFG0M/ref=ssq_hps_bw_g21-ir03?pf_rd_m=ATVPDKIKXODER&pf_rd_s=center-3&pf_rd_e=gxwy42FH2K03Y7BMWQNM&pf_rd_t=101&pf_rd_p=1328901542&pf_rd_i=165379



Analyzing Requirements

- **Analyzing** requirements is the process of examining and understanding the requirements of a system.
- It involves identifying dependencies, conflicts, and inconsistencies in the requirements.
- It is a critical activity that ensures that the requirements are complete, consistent, and correct.

Meeting with the client



Analyzing Requirements

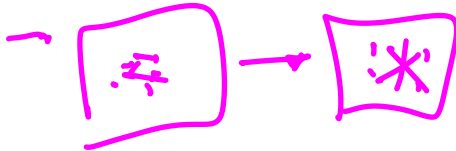
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Documenting Requirements

- **Documenting** requirements is the process of writing and organizing the requirements of a system.
- It involves creating documents, diagrams and models that describe the requirements in a clear and concise way.

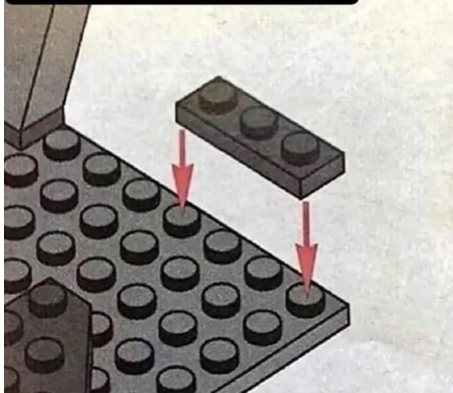
→ what is
→ how to use



Everyone hates writing documentation

**En la documentación
está todo bien explicado**

La documentación:



*bad
quality*



Validating Requirements

- **Validating** requirements is the process of ensuring that the requirements are correct and complete
- It involves reviewing the requirements with stakeholders to verify that they meet their needs.



Unclear Understanding of Requirements



Dad Jokes
@Dadsaysjokes

My dad told me his password is:
MickeyMinnieGoofyDonaldPlutoHuey
LouieDeweyDublin.

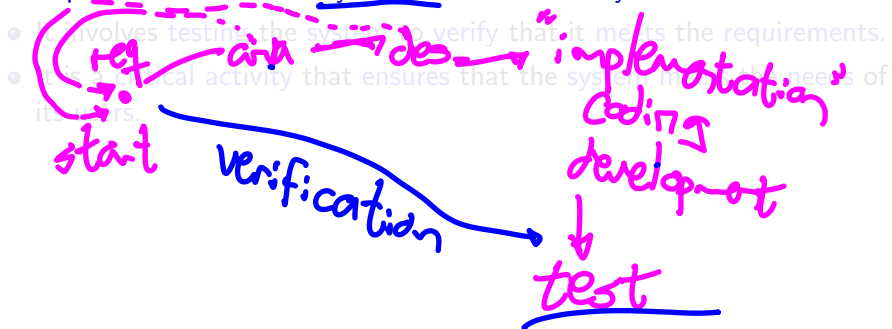
Because he was told his password
had to contain 8 characters and at
least one Capital.

Password



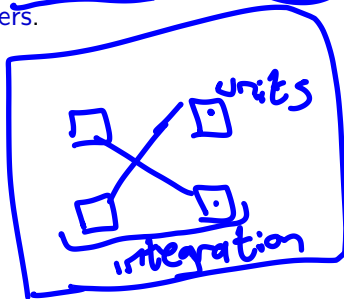
Verifying Requirements

- **Verifying** requirements is the process of ensuring that the requirements are correctly implemented in the system.



Verifying Requirements

- **Verifying** requirements is the process of ensuring that the requirements are correctly implemented in the system.
- It involves testing the system to verify that it meets the requirements.
- It is a critical activity that ensures that the system meets the needs of its users.



system → acceptance



Typical Mistakes when Testing

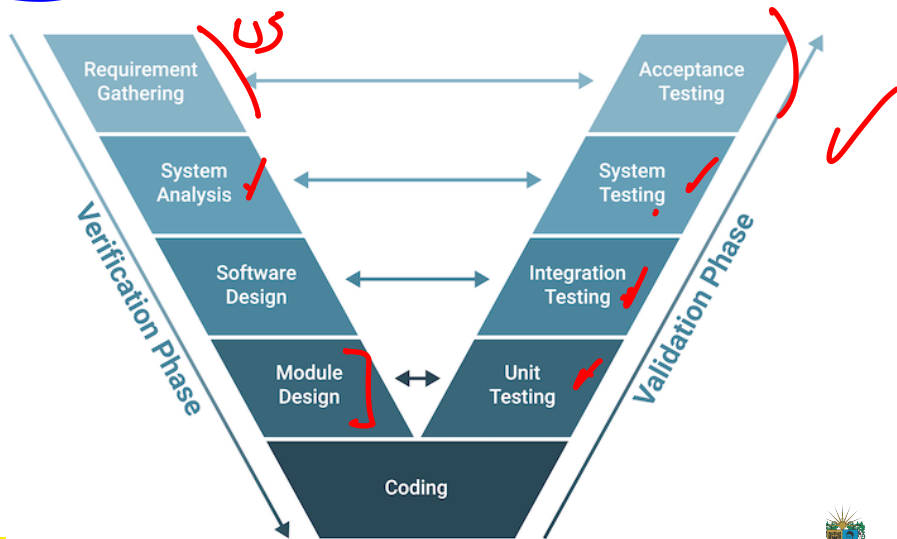
Disturbing Chinese calorie app...



	カシューナッツ (cashew)	1 粒	9 kcal
	ジャムパン (Pomeranian)	1 個	327 kcal



V-Model in SDLC



Outline

User stories
↓
written

functional requirements

User interaction

1 Requirements Engineering

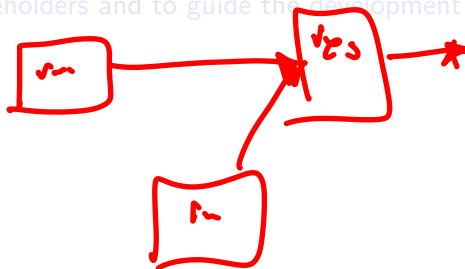
2 Design & Process

non-functional requirements
↙
quality



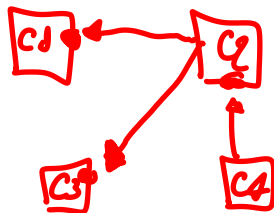
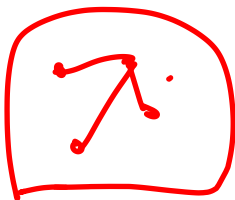
Conceptual Design

- Once an initial set of **requirements** is defined, the next step is to create a **conceptual design** of the system. *non-technical*
- Conceptual Design** is a **high-level design** that defines the **structure** and **behavior** of the **system**. It is achieved through the recognition of the appropriate **components**, **connections**, and **responsibilities**.
- The **conceptual design** is used to **communicate** the **vision** of the system to **stakeholders** and to **guide** the **development** of the system.



Conceptual Design

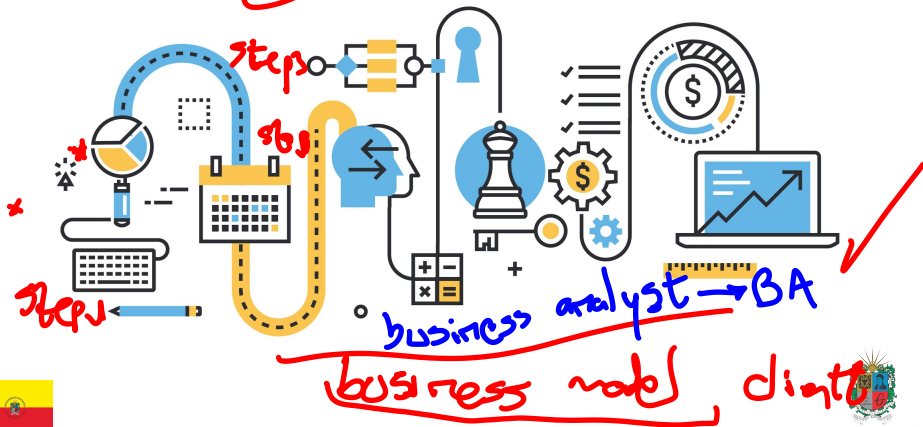
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Process Definition

→ algorithm

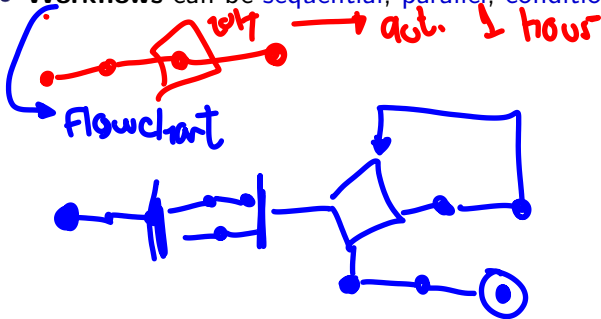
- A **Process** is a **series** of steps or **actions** taken to achieve a particular goal.
- **Processes** are used to **organize** and **manage** work.



Workflows

process - algorithm

- A **Workflow** is a series of tasks that are performed in a specific order to achieve a goal.
- **Workflows** are used to automate and optimize business processes.
- **Workflows** can be sequential, parallel, conditional, or repetitive.



Process Models

subsystem

graphic

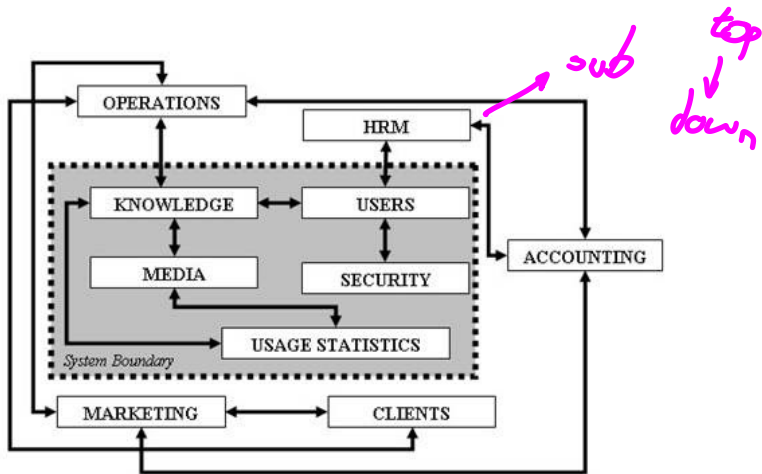
- A Process Model is a representation of a process that shows the sequence of steps and the relationships between them.
- **Process models** are used to analyze, design, and improve processes.
- Examples of process models include flowcharts, data flow diagrams, activity diagrams, business process model and notation (BPMN), Petri nets, state diagrams, among others.



*roles**
tasks
Flow



System Schema Example: Company Structure



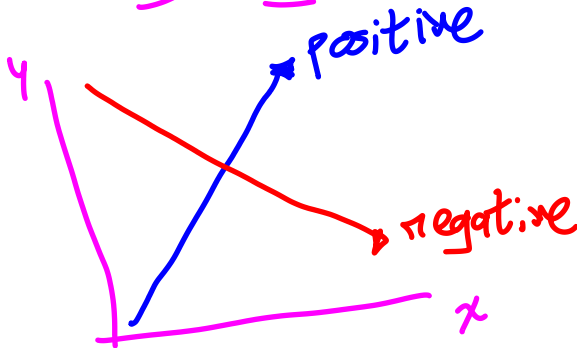
Causal Loops

- A **Causal Loop** is a diagram that shows the relationships between different variables in a system.
- Causal loops are used to analyze and understand the dynamics of a system.
- Causal loops can be positive or negative.

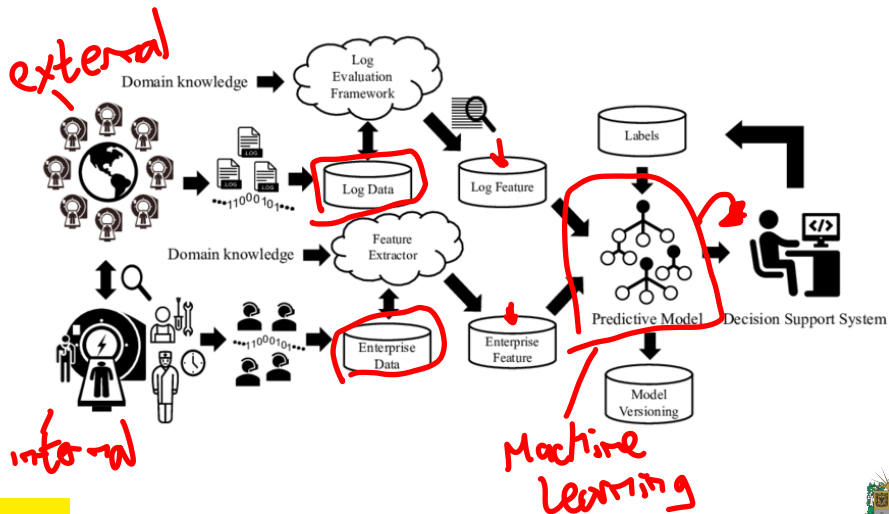


Causal Loops

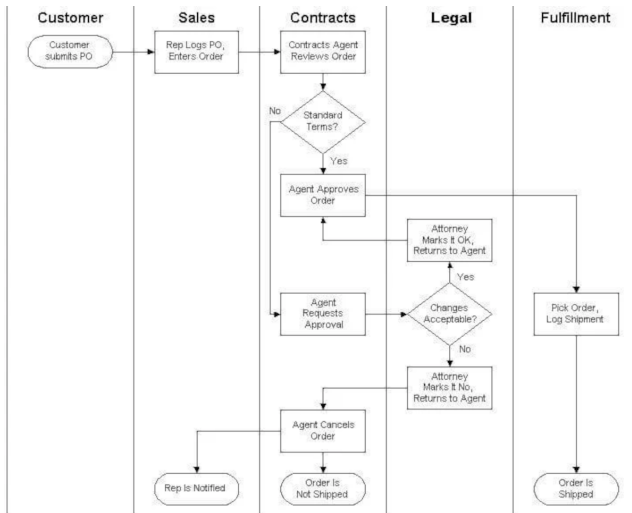
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System Schema Example: Processing Pipeline



Business Process Model and Notation (BPMN)



Technical Design

- Once the **conceptual design** of the system is defined, the next step is to create a **technical design** of the system.
- **Technical Design** is a **detailed design** that defines the architecture, components, and interfaces of the system.
- The **technical design** is used to **guide the development** of the system and to **communicate the implementation details to developers**.

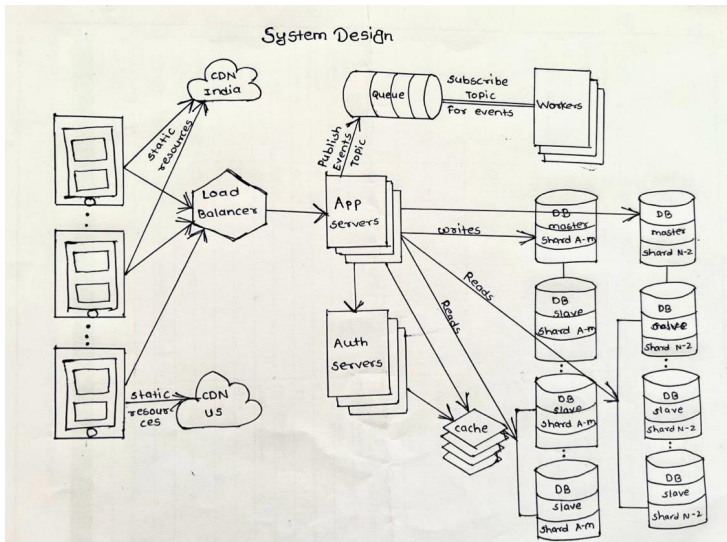


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Systems Design Applied to Software Architectures



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Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/systems-analysis>

